

Hoang Minh Thanh

Research Engineer

 [hmthanh.github.io](https://github.com/hmthanh)

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 [OCID](#)

 [Google scholar](#)

★ Keypoint

- **Experience AI Model Evaluation & Dataset Analysis:** Evaluated AI models performance [GENEA Leaderboard](#) [🔗](#).
- **Proficiency in Machine Learning and Deep Learning Frameworks:** Hands-on development and integration of OTS engines using PyTorch, focusing on translational research for platforms analyzing internet content, as demonstrated in [DeepGesture](#) [🔗](#).
- **Experience in Multi-Modal Large Language Models:** Applied MM LLMs to evaluate and adapt tools for toxic content detection, contributing to stakeholder-specific applications in collaborative environments.
- **Strong Collaboration and Ownership in Interdisciplinary Settings:** Proven ability to engage with research, industry, and government stakeholders at events, while taking ownership of tasks and providing constructive feedback for team growth.

🎓 Education

AS	University of Science - VNUHCM, Information Technology	2012 – 2015
BS	University of Science - VNUHCM, Computer Science	Sept. 2018 – Sept. 2020
	• GPA: 3.3/4.0 • Thesis: GCAT - Link Prediction in Knowledge Graph, [Report], [Code]	
MSc	University of Science - VNUHCM, Computer Science	Oct. 2021 – Dec. 2024
	• GPA: 3.44/4.0 • Thesis: OpenHuman: A conversational gesture synthesis system based on emotions and semantics, [Report], [Code]	

📖 Publications

DeepGesture: A conversational gesture synthesis system based on emotions and semantics [arXiv] [Homepage]	Jul. 2025
Towards a GENE Leaderboard [arXiv]	Oct. 2024
Graph Collaborative Attention Network for Link Prediction in Knowledge Graphs [arXiv] [Homepage]	Oct. 2020

💼 Experience

Rakumo Inc, Software Engineer	Jul. 2019 – Apr. 2020
• Developed a Django-based web application with integrated DocuSign.	
FPT Software, AWS Data Engineer	Oct. 2020 – Nov. 2021
• CI/CD data pipeline: implemented and modified real-time and batch data pipelines on AWS using Kinesis, Lambda, S3, Glue, StepFunction, SNS, ..etc.. • Cox Automotive (onsite-offshore): Collaborated with international teams in Agile	
VNG Corp, Software Engineer	Oct. 2021 – May 2024

ZDN Team (Zalo Content Delivery Network): caching service serve millions request

- Implement scheduled `ThreadPool` for monitoring stats of server execution.
- Built real-time server status with visualization via `amchart`, `p5.js` in `NextJS`.
- Integrate outlier detection to prevent DDoS attacks in the caching server.

OpenHuman, Creator

May 2024 – Present

- Join GENE Research Team ([GENEA Leaderboard](#), [HEMVIP](#)): Designed and built a full-stack, production-ready leaderboard system to evaluate multiple AI gesture generation models, integrating user study workflows via Prolific.

</> Pet Projects

MillionScope ([Code](#)) (Fullstack self-made)

[millionscope.com](#) [↗](#)

- Custom Reasoning LLM Chat with ChatGPT-Style Interface, Streaming Responses, and Result Annotations.

SurveyWithCode ([Code](#)) (In progress)

[surveywithcode.dev](#) [↗](#)

- Remake `PapersWithCode` using `Nextjs` and `d3.js`

Three.js realistic face ([Code](#)): Loaded and scaled facial OBJ parts in Blender, applied edited texture maps (albedo, roughness, specular, clearcoat) and tone mapping

[renderengine.pages.dev](#) [↗](#)

Unity Game Pucca Runner ([Code](#)) Pathfinding with a 2D hash map and Dijkstra's algorithm; runner moves toward the player and triggers collision on boundary breach

[Pucca Runner](#) [↗](#)

Others: [Morph target animation](#), [MetaPet](#) ([metapet.vercel.app](#)), [cortexpod.com](#), [pithagon.com](#)

🌟 Certifications

- [Critical Thinking Certification 2014](#) [↗](#), [Emotion Recognition Certification 2019](#) [↗](#), [Summer ML](#) [↗](#)
- Coursera: [Design Pattern](#) [↗](#), [TensorFlow Developer Professional](#) [↗](#) 2021
- **AWS:** [Solutions Architect - Associate](#) [↗](#), [Machine Learning - Specialty](#) [↗](#); [Solution Architect](#) [↗](#) 2021
- [Solution Architect](#) [↗](#) 2021

💻 Technologies

- **AI/ML & Model Evaluation:** Machine Learning Frameworks (e.g., `PyTorch`, `TensorFlow`), Multi-Modal Large Language Models, Model Evaluation and Testing, `MLflow`, `Weights & Biases`.
- **Programming Languages:** `Python`, `Java`, `C/C++`
- **Development Tools:** `Jupyter Notebook`, `VS Code`, `Git`